

1. PROBLEM & OBJECTIVES

- Tea estate productivity is affected by labour shortages, worker welfare challenges, leaf diseases, changing weather and quality assessment.
- Traditional manual decision-making can delay access to useful operational information.
- Objective:** integrate AI, ML, DL, and IoT into one decision-support platform for estate managers.



Research gap

Existing studies commonly address individual tasks; this system combines four management functions in one platform.

2. FOUR INTELLIGENT MODULES



LABOR & WELFARE

- AI task allocation
- HR + SpO₂ monitoring
- Stress prediction

LightGBM • 82%



LEAF DISEASE

- Image detection
- Confidence score
- Treatment recommendation

YOLOv8s • 96.1% mAP@0.5



WEATHER & CLIMATE

- Temperature forecast
- Disease-risk prediction
- Estate recommendations

GBR + Random Forest • 89%

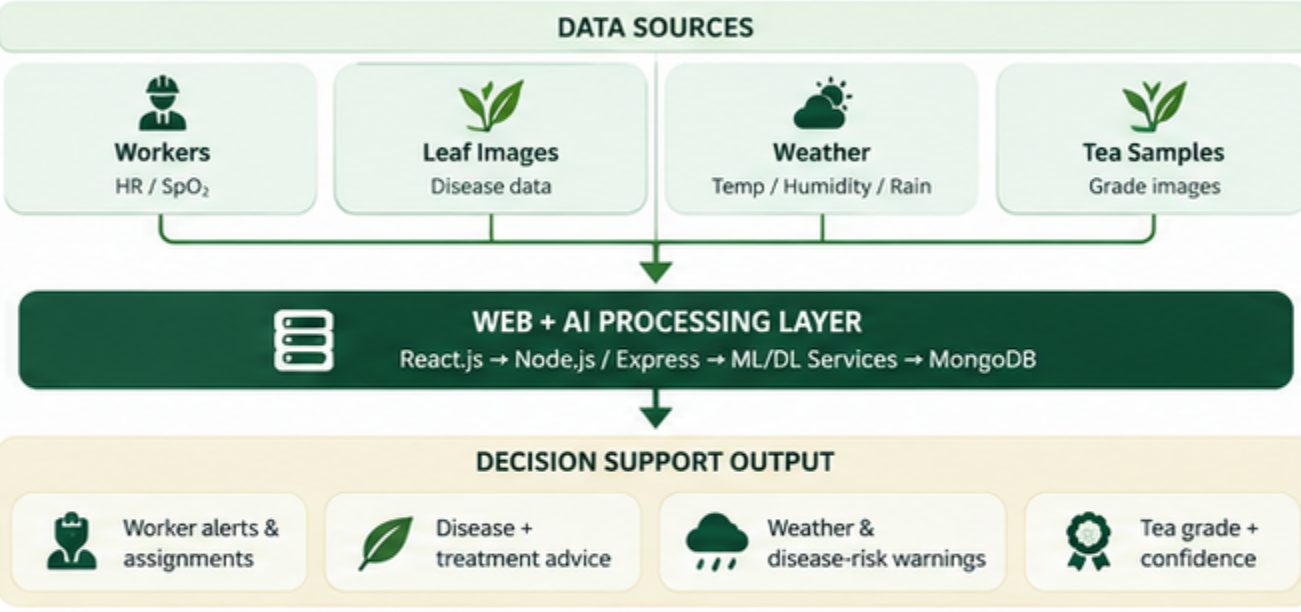


TEA SORTING & GRADING

- Processed-tea image analysis
- Automatic grade classification
- Quality assessment

ConvNeXtV2-Tiny • 99.7% F1

3. SYSTEM ARCHITECTURE

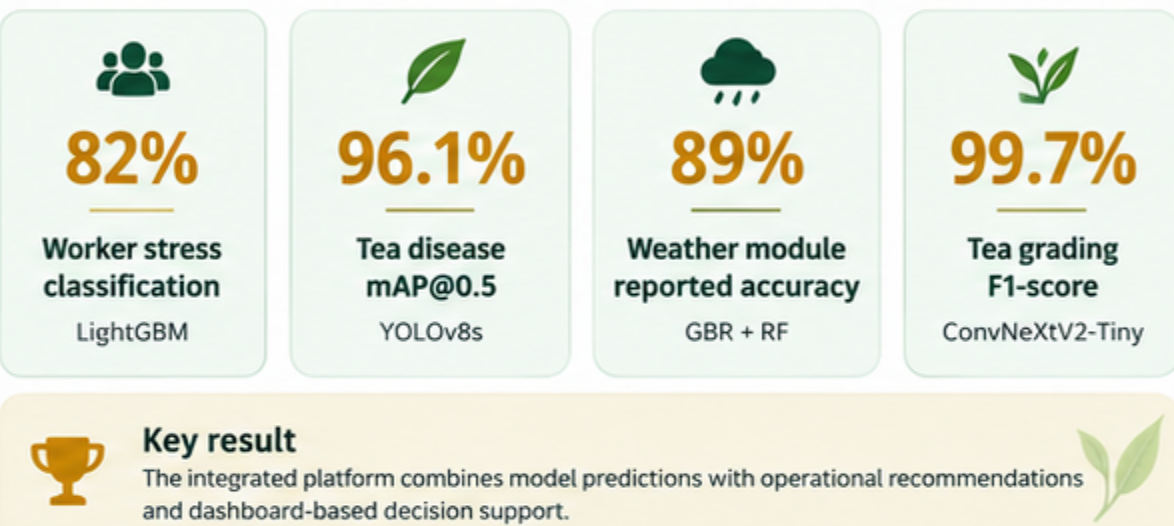


4. TECHNOLOGY & METHODOLOGY

- Frontend:** React.js web application.
- Backend:** Node.js and Express.js; database: MongoDB.
- Models:** LightGBM, YOLOv8s, Gradient Boosting, Multi-output Random Forest, ConvNeXtV2-Tiny.
- IoT:** wearable HR and SpO₂ monitoring.
- Weather:** historical and real-time meteorological data.



5. EXPERIMENTAL RESULTS



6. CONTRIBUTION & CONCLUSION

- Unified platform for labour, disease, weather, and tea-quality decision support.
- Real-time worker monitoring with AI-based prediction.
- Early disease-risk information and treatment-oriented guidance.
- Automated tea grading using deep learning.
- Data-driven and climate-aware estate management.

Conclusion

AI, ML, DL, and IoT can be integrated into a single web-based platform to support timely tea-estate management decisions.



FUTURE WORK

Expand datasets • Improve models • Add IoT devices • Develop mobile version

Based on the TeaGuard research manuscript submitted to ICAC 2026.



RESEARCH TEAM

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